

Due on monday, October 22rd, before 5pm.

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1. Matrices (or operators) might not commute with each other. This means, if $\hat{A}$ and $\hat{B}$ are matrices (or operators), $\hat{A}\hat{B}$ might be unequal to $\hat{B}\hat{A}$. We define the **commutator** of two matrices (or two operators) $\hat{A}$ and $\hat{B}$ as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

Sometimes operators acting on functions are written with a hat, as I've done here. Operators acting on finite vectors, i.e., matrices, are usually written without a hat. Let's use the hat notation for this problem.

- (a) [**2 pts.**] Show that $[\hat{B}, \hat{A}] = -[\hat{A}, \hat{B}]$.
 - (b) [**SELF**] Show that $[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]$.
 - (c) [**4 pts.**] Show that $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$.
2. (a) [**0 pts.**] Look up the three Pauli matrices, denoted as σ_x , σ_y , σ_z , and write them down, (Each one is a 2×2 matrix.)
- (b) [**4 pts.**] Calculate $[\sigma_x, \sigma_y]$ and express it in terms of σ_z .
- (c) [**2 pts.**] Look up (or calculate or guess correctly) $[\sigma_y, \sigma_z]$ in terms of σ_x , and $[\sigma_z, \sigma_x]$ in terms of σ_y . No need to show calculations; just report the results.
- (d) [**7 pts.**] Find the eigenvalues and (normalized) eigenvectors of σ_x .
- (e) [**SELF**] Find the eigenvalues and (normalized) eigenvectors of σ_z .
- (f) [**SELF**] Is it easier to find eigenvalues & eigenvectors if the matrix is diagonal? Why?
3. (a) In Charles Nash's notes (linked on module webpage): Read Section 5 of Chapter III, on 'Expectation Values'. Begins on page 42.
- (b) [**0 pts.**] Define the expectation value of an observable represented by operator $\hat{A}$, when the system state is $|\psi\rangle$.
- (c) [**6 pts.**] The state $|\phi\rangle$ is an eigenstate of $\hat{A}$, with the corresponding eigenvalue being λ . What is the expectation value of $\hat{A}$ in this state? (Prove/derive your answer.)

4. For a spin-1/2 system, the operators for the components of spin are

$$S_x = \frac{\hbar}{2}\sigma_x \quad S_y = \frac{\hbar}{2}\sigma_y \quad S_z = \frac{\hbar}{2}\sigma_z$$

- (a) **[6 pts.]** Find the eigenvalues and eigenvectors of S_z . How are they related to the eigenvalues and eigenvectors of σ_z ?
- (b) **[6 pts.]** If our two-level system is in the state $|\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, then find the expectation values of S_z and S_x .
- (c) **[SELF]** Which one of the expectation values could you have guessed?
- (d) **[SELF]** Find the commutation relations between the operators S_x , S_y and S_z . (You may use the results for commutators of Pauli matrices.)
5. Photoelectric effect: When electromagnetic radiation of frequency 2×10^{15} Hz shines on a sample of gold, the emitted electrons have a kinetic energy of 3.0 eV. When this frequency is increased to 10^{16} Hz, the electron kinetic energy is 36.1 eV.
- (a) **[0 pts.]** How much is 1 eV of energy, in SI units?
- (b) **[6 pts.]** Pretending that we do not know the value of Planck's constant (h), use the above data to estimate h in SI units (J-s). Use Einstein's energy conservation equation for the photoelectric effect.
- (c) **[SELF]** Use the above data to find the work function of gold, in eV and also in Joules. (You don't necessarily need the value of h for this, but if you want you can use $h = 6.626 \times 10^{-34}$ J-s.) What does the work function physically represent?
6. **[7 pts.]** Look up the properties of the Dirac delta function. If a is a positive real number, write down the values of the following integrals.

$$\begin{aligned} \text{(a)} \quad & \int_{-a}^a dx \, \delta(x) & \text{(b)} \quad & \int_a^{2a} dx \, \delta(x) & \text{(c)} \quad & \int_{-a}^a dx \, \delta(x)f(x) \\ \text{(d)} \quad & \int_a^{2a} dx \, \delta(x)f(x) & \text{(e)} \quad & \int_{-a}^a dx \, \delta(x-2a) \\ \text{(f)} \quad & \int_{-a}^a dx \, \delta(x-2a)f(x) & \text{(g)} \quad & \int_0^{3a} dx \, \delta(x-2a)f(x) \end{aligned}$$

Hint: plotting the integrand in the region within the limits of integration is often helpful.